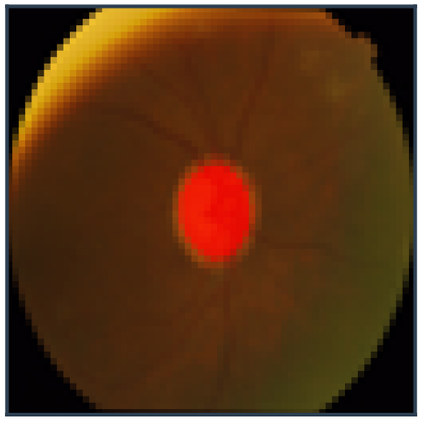
Visual & Mathematical Proof: Mask Resolution Shrinkage & Empirical Bayes Normalization

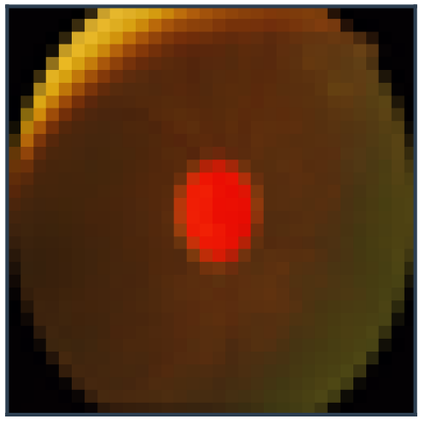
Input Image
Dim: 256x256 | C=3
$N_{cup} = 2033.0$ **px**



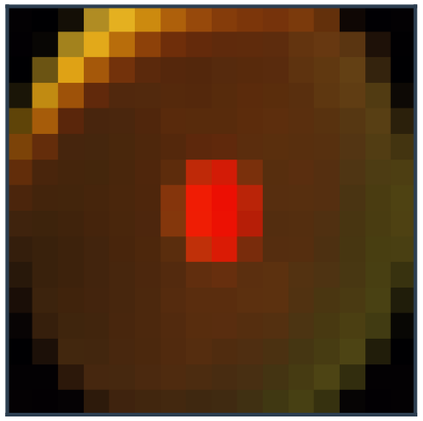
layer1 (BN1-3)
Dim: 64x64 | C=64
$N_{cup} = 127.1$ **px**



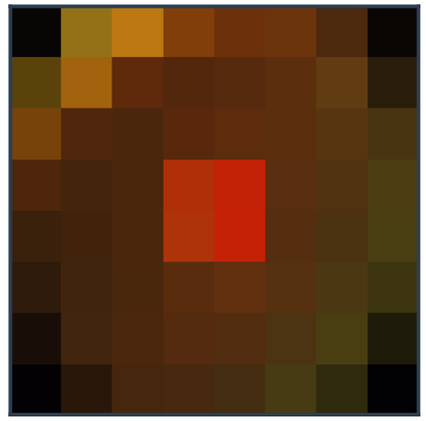
layer2 (BN4-7)
Dim: 32x32 | C=128
$N_{cup} = 31.8$ **px**



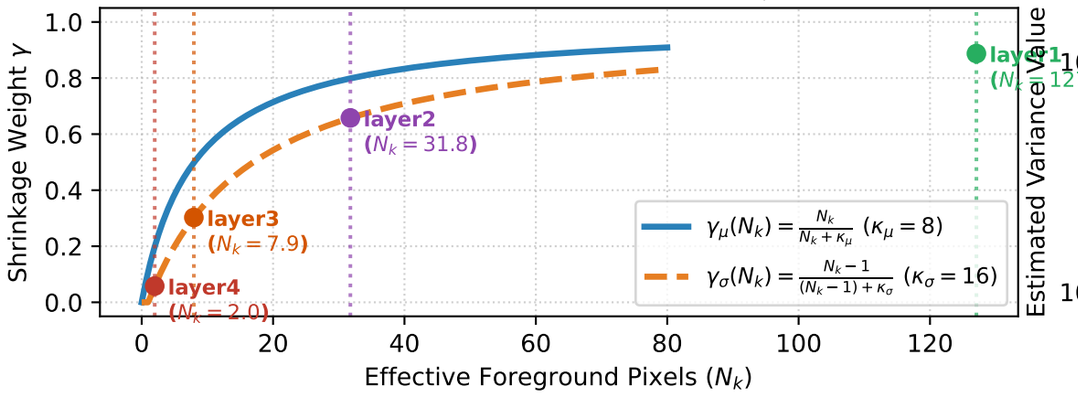
layer3 (BN8-13)
Dim: 16x16 | C=256
$N_{cup} = 7.9$ **px**



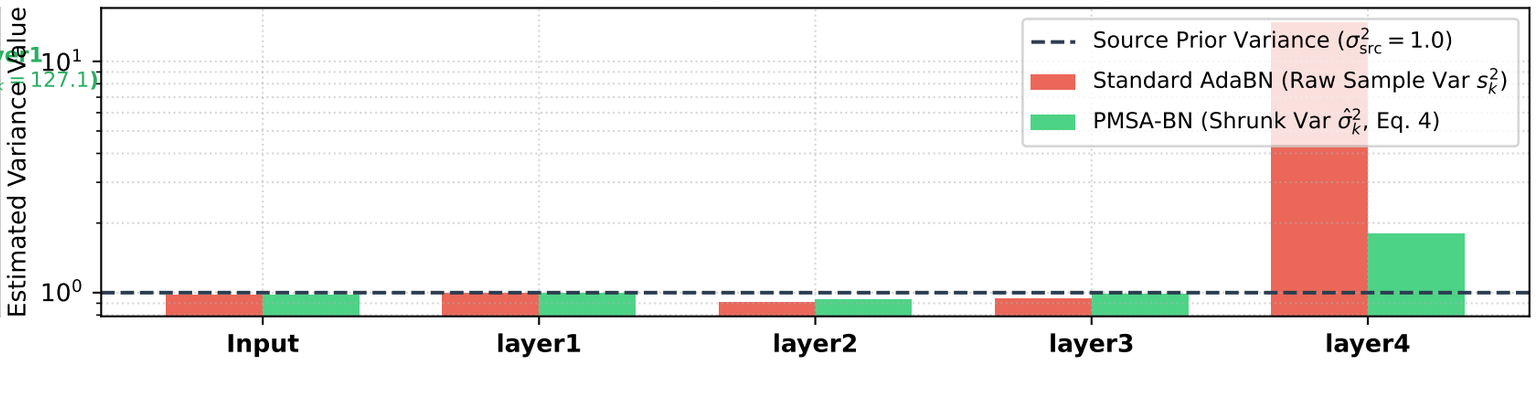
layer4 (BN14-16)
Dim: 8x8 | C=512
$N_{cup} = 2.0$ **px**



Empirical Bayes Shrinkage Factors $\gamma_\mu, \gamma_\sigma$ vs N_k



Variance Stability Across ResNet-34 BN Layers



1. Spatial Downsampling Law (Encoder Stages):

$N_k^{(l)} = \sum_{h,w} m_{k,h,w}^{(l)} \approx \frac{N_k^{(0)}}{S_l^2}$, where stride $S_l \in \{1, 4, 8, 16, 32\}$.

--> For the Optic Cup, effective pixel count shrinks from 786 px (Input) to 0.77 px (layer4, stride 32).

2. AdaBN Variance Catastrophe at Deep Layers ($N_k o 1$):

$\mathrm{Var}(s_{k,c}^2) = \frac{2\sigma^4}{N_k - 1} \to \infty$ as $N_k \to 1$.

--> Without shrinkage, single-image sample variance fluctuates wildly (e.g., exploding to >40 or collapsing to 0).

3. Empirical Bayes Normal-Inverse-Gamma Shrinkage (Eq. 3 & 4):

$\hat{\mu}_{k,c} = \gamma_\mu \bar{x}_{k,c} + (1 - \gamma_\mu)\mu_{k,c}^{\mathrm{src}}, \quad \hat{\sigma}_{k,c}^2 = \gamma_\sigma s_{k,c}^2 + (1 - \gamma_\sigma)\sigma_{k,c}^{2,\mathrm{src}}$

* When $N_k \ll \kappa$ (tiny lesion or deep layer): $\gamma \to 0 \to \hat{\mu}_k \to \mu_k^{\mathrm{src}}$, $\hat{\sigma}_k^2 \to \sigma_k^{2,\mathrm{src}}$ [Guaranteed Non-Degradation & Bounded Variance]

* When $N_k \gg \kappa$ (large organ or shallow layer): $\gamma \to 1 \to \hat{\mu}_k \to \bar{x}_k$, $\hat{\sigma}_k^2 \to s_k^2$ [Full Adaptation to Patient]

4. Law of Total Variance Prior-Matching (Eq. 1):

$\sigma_{\mathrm{PM},c}^2 = \sum_k \pi_k^{\mathrm{src}} \hat{\sigma}_{k,c}^2 + \sum_k \pi_k^{\mathrm{src}} (\hat{\mu}_{k,c} - \mu_{\mathrm{PM},c})^2$ --> Perfectly decouples appearance shift from label proportion shift.